



University of Colombo, Sri Lanka

UCSC University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester II— UCSC AY19 [held in March/April/May 2023]

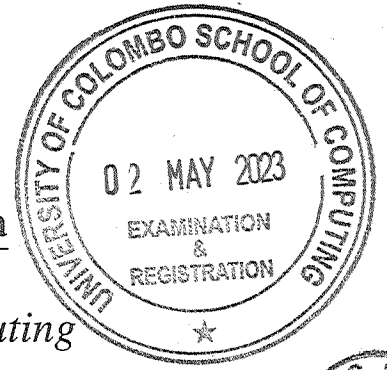
SCS1214 — Operating Systems I

(2 Hours)

Answer All Questions

Number of Pages = 10

Number of Questions = 4



234

To be completed by the candidate

Index Number

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Important Instructions to candidates:

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Write your index number **CLEARLY** on each and every page of this Question paper.
- This paper consists of 4 questions in 10 pages (including the Cover Page).
- Answer **ALL** questions.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

1	
2	
3	
4	
Total	

Index Number

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1. (a). A code segment of the virtual machine used for the SCS1214 assignments is given bellow.

```
....
def mod(opr):
    reg['acc']=reg[opr[0]] % reg[opr[1]];
    reg['pc']=reg['pc']+1;

def call(opr):
    reg['sp']=reg['sp']+1;
    memory[reg['sp']]= X;
    reg['pc']=int(opr[0]);

def ret(opr):
    reg['pc']=memory[reg['sp']];
    reg['sp']= Y ;

def push(opr):
    reg['sp']=reg['sp']+1;
    memory[reg['sp']]=reg[opr[0]];
    reg['pc']=reg['pc']+1;

def pop(opr):
    reg[opr[0]]= Z ;
    reg['sp']=reg['sp']-1;
    reg['pc']=reg['pc']+1;

def jmp(opr):
    reg['pc']=int(opr[0]);
....
```

i. What is X?

[3 marks]

--

ii. What is Y?

[3 marks]

--

iii. What is Z?

[3 marks]

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Index Number

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- (b). Ajantha has compiled and run the following programme on a machine running the Linux operating system. The line numbers are given in the left.

```
1 int x[4096][4096];
2
3 int main() {
4 int i, j;
5
6 for(i=0; i < 4096; i++)
7     for(j = 0; j < 4096; j++)
8         x[j][i]=125;
9 return 0;
10 }
```

Ajantha used a debugger and discovered that $x[0][0]$ is at the memory location 1109667776 and $x[1][0]$ is at the memory location 1109684160. The page size used by the system is 4096 bytes and the size of an integer is 4 bytes.

Assume that the operating system only allocates 100 pages for a process and uses demand paging. Also assume that the process executing the above programme completed without any interruptions or context switches.

- i. Which memory location is assigned 125 when $i=0$ and $j=2$

[4 marks]

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- ii. How many page faults are caused by the nested loop in the above programme?

[4 marks]

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- iii. Ajantha realized that by changing one line of the programme he can reduce the number of page faults drastically without changing the semantics of the code. Write that changed line.

[4 marks]

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- iv. How many page faults were generated after the above (1(b)iii) change?

[4 marks]

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Index Number

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2. (a). i. Following programme produced two numbers as output.

```
int main()
{
    printf("%d\n", fork());
}
```

One number is 73368. What is the other number? Assume that `fork()` completed successfully.

[4 marks]

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- ii. What is the output of the following programme? Assume that all `fork()` calls completed successfully.

```
int main()
{
    int x=0;
    if (! fork()) x++;
    if (! fork()) x++;
    if (x % 2) printf("%d\n", x);
}
```

[5 marks]

--

- iii. `fork()` creates a child process by copying the memory content of the parent. However, bulk of the copying can be avoided and the necessary copying can be delayed. Describe the technique used to achieve this?

[5 marks]

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Index Number

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- (b). A system uses page tables with x entries. The size of a page is p bytes. How many bits are used for the virtual addresses in this system?

[5 marks]

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- (c). The structure of the producer process of the bounded buffer problem is given belllow.

```
do {  
    ...  
    wait(empty);  
    wait(mutex);  
    /* add the produced item to the buffer */  
    signal(mutex);  
    signal(full);  
} while (true);
```

The structure of the consumer process is given bellow, but with place holders X,Y, P, Q for actual parameters.

```
do {  
    wait(X);  
    wait(Y);  
    /* remove an item from the buffer*/  
    signal(P);  
    signal(Q);  
    ...  
} while (true);
```

- i. What is X?

[2 marks]

--

- ii. What is Y?

[2 marks]

--

- iii. What is Q?

[2 marks]

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Index Number

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3. (a). What is multi-programming and what is its purpose?

[3 marks]

--

- (b). What is **CPU utilization** ? Define it.

[2 marks]

--

- (c). What is the purpose of **CPU scheduling**?

[3 marks]

--

- (d). What is the data structure that contains all the details of a process that is waiting to be scheduled on CPU?

[2 marks]

--

Index Number

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(e). What is the purpose of a dispatcher and what are its features?

[3 marks]

--

(f). What is meant by **processor affinity**? What types of affinity is crucial for time-critical real-time systems?

[2 marks]

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(g). List two (2) limitations each for single-level and two-level directory structures.

[5 marks]

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Index Number							
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(h). What is the task of the memory management unit (MMU) ?

[3 marks]

--

(i). What is demand paging ?

[2 marks]

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Index Number

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4. (a). List the main steps of basic page replacement.

[10 marks]

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- (b). You are given a system that consists of 32-bit addressing. Each data block is 1KB and the pointer size is 8 bytes in size. What is the maximum file size that can exist in your file system?

Assume that there are 15 direct pointers to data blocks, 5 single indirect pointers, 3 double indirect pointers, 2 triple indirect pointers and 1 quadruple indirect pointer defined in the file system.

[10 marks]

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Index Number

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(c). What is Belady's anomaly? Which page replacement algorithm(s) be used to prevent it?

[3 marks]

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(d). What is the difference between Process VMs and System VMs? Explain your answer in terms of virtualizing interface.

[2 marks]

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